TEST CODE **02107032**

FORM TP 2013145

MAY/JUNE 2013

C A R I B B E A N   E X A M I N A T I O N S   C O U N C I L

CARIBBEAN ADVANCED PROFICIENCY EXAMINATION®

BIOLOGY

UNIT 1 – Paper 032

ALTERNATIVE TO SCHOOL-BASED ASSESSMENT

2 hours

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of THREE questions. Answer ALL questions.
2. Write your answers in the spaces provided in this booklet.
3. You may use a silent non-programmable calculator.
4. Candidates are advised to use the first 15 minutes for reading through this paper carefully.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

Please begin Question 1 FIRST.

1. (a) (i) You are required to carry out a simple investigation to determine the sucrose concentration which is isotonic to the potato tuber cells.

You are provided with the following materials:

- Potato tuber cylinders
- 0.1 M, 0.3 M, and 0.6 M sucrose solution
- Distilled water

PROCEDURE

1. Place the 0.1 M, 0.3 M, and 0.6 M sucrose solution and distilled water in four beakers labelled A, B, C and D, as shown in the following table.

Beaker	**Concentration of Solution**
A	0.1 M
B	0.3 M
C	0.6 M
D	Distilled water

2. Remove any excess water from the potato cylinder using the paper towel. Measure the **initial** length of the potato cylinders to the nearest cm and record the average length in Table 1 on page 3.

3. Place three cylinders into each of the beakers labelled A, B, C and D **and let stand for one hour**.

4. Proceed to Questions 2 and 3.

5. After one hour, remove the potato cylinders from the beakers and blot to remove excess solution.

6. Measure the length of EACH potato cylinder in Beaker A and calculate the average length. Record your answer in Table 1 on page 3.

7. Repeat Step 6 for Beakers B, C and D.

8. For EACH solution, calculate the average change in length and record your findings in Table 1 on page 3.

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TABLE 1: ____

____

Length (cm)	**A 0.1 M**	**B 0.3 M**	**C 0.6 M**	**D Distilled Water**
FINAL average length (cm)				
INITIAL average length (cm)				
Change in average length (cm)				

[6 marks]

(ii) Suggest a specific aim for your experiment.

____

____

[1 mark]

(iii) Suggest an appropriate title for Table 1. Write your answer on the lines provided above the table. **[1 mark]**

(iv) Account for the change in length of the potato cylinders at 0.1 M and 0.6 M.

0.1 M ____

____

____

0.6 M ____

____

____

[2 marks]

(v) Describe TWO ways of using this data to estimate the osmolarity of the potato tuber.

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[4 marks]

(b) Figure 1 is an electron micrograph of organelles in a eukaryotic cell.

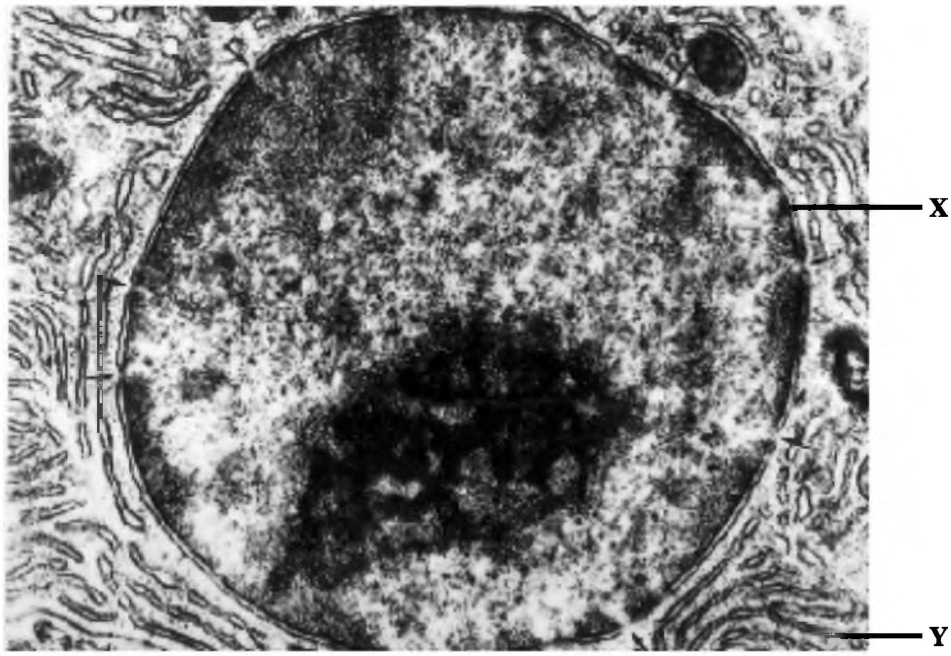

Figure 1. Electron micrograph of a eukaryotic cell

http://online-media.uni-marburg.de/histologie/introhis/HIS/txt/tacsem/tac01_sem.htm

Identify the structures labelled X and Y in Figure 1.

X: ____

Y: ____

[2 marks]

Total 16 marks

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2. (a) In an experiment, a pigeon pea plant with green, round seeds and a pigeon pea plant with white, wrinkled seeds were crossed. All the F_1 plants had green, round seeds. The F_1 plants were then selfed. Table 2 is an incomplete table for the results of the F_1 cross.

(i) Complete Table 2 by writing the missing values for the F_2 progeny, and the expected phenotypic ratio for the cross.

TABLE 2: RESULTS OF F_1 CROSS

Phenotype/Seed Colour and Shape	**Number of Progeny**	**Expected Phenotypic Ratio**
Green round	780	
Green white	148	
White round		
White wrinkled		
Total	1136	

[3 marks]

(ii) Explain, giving TWO reasons, why all of the **F_1** progeny were phenotypically alike.

Reason: ____

____

Reason: ____

____

[2 marks]

(iii) In the space provided below, construct a Punnett diagram for the test cross of a F_1 plant. Use the symbol G for the colour of the seeds and R for the shape of the seeds. (Include in your diagram an indication of the genotypic ratios for phenotypic categories.)

[6 marks]

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(b) You are provided with a fresh slide preparation of onion root tip squash. In the box below, draw a cell undergoing metaphase of mitosis.

[5 marks]

Total 16 marks

3. (a) Figure 2 is a photomicrograph of a reproductive cell.

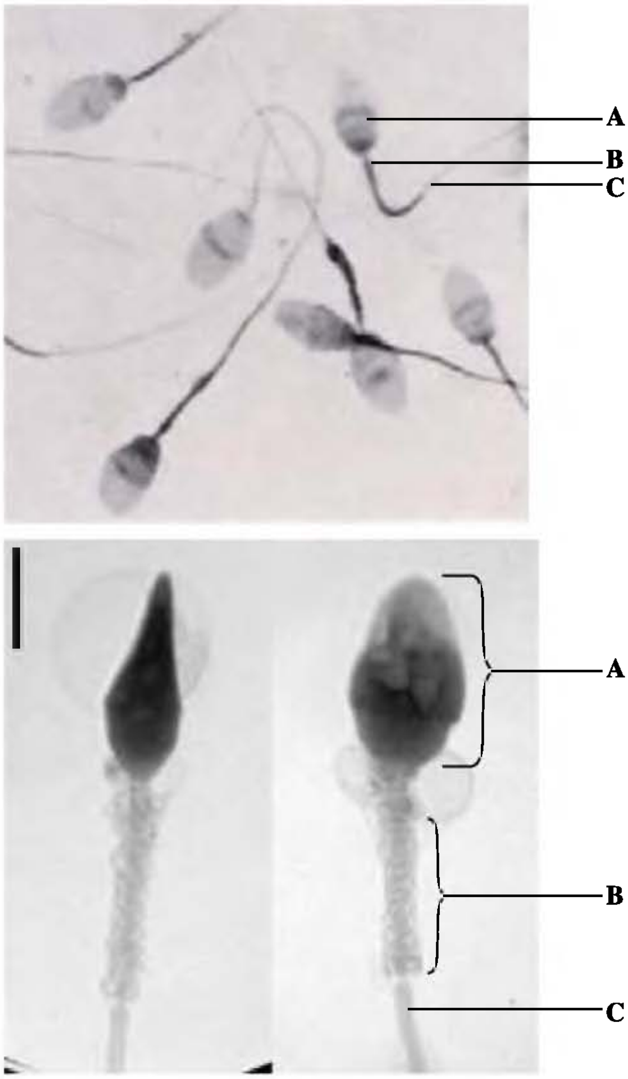

Figure 2. Photomicrograph of a reproductive cell

http://www.phys.au.dk/~jabraham/sperm98.gif

(i) Name the reproductive cell.

____

[1 mark]

(ii) Identify the structures labelled A, B and C in Figure 2.

A: ____________________________________

B: ____________________________________

C: ____________________________________

[3 marks]

(iii) Comment on the function of the structure labelled B.

____

____

____

____

[3 marks]

(b) Table 3 shows data on the effectiveness of contraceptive methods, expressed as a percentage of women likely to become pregnant while using the method for one year.

TABLE 3: PREGNANCY RATES FOR CONTRACEPTIVE METHODS

Contraceptive Method	**Rate of Pregnancy (expressed as a %)**
Male sterilization	0.1
Female sterilization	0.5
Pill (combined progesterone/estrogen)	0.15
Intrauterine device	0.6
Male condom	2.0
Female condom	5.0
Spermicide	18.0
Natural method (*calendar, temperature, cervical mucus*)	9.0

(i) On the grid provided on page 9, construct a histogram for the data given in Table 3. **[4 marks]**

(ii) Determine the most effective and the least effective method of contraception. Cite the data used to justify your answer.

Most effective method: ____

Least effective method: ____

[3 marks]

GO ON TO THE NEXT PAGE

(iii) Identify ONE method of contraception from Table 3, **other than** the male condom, which may give protection from HIV infection. Give ONE reason for your answer.

[2 marks]

Total 16 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.